

3D Mapping Based on LiDAR Signal Processing

Illia Novosad

Czech Technical University in Prague

June 17, 2026



- 1 Introduction
- 2 Platform Design
 - Existing Amateur Solutions for 3D Scanning
 - Point Cloud Visualisation at Different Tilt Angles
 - Dependence of Scan Spacing S_{scan} on Tilt Angle θ
 - Effect of Reduction Ratio R on Scan Spacing S_{scan}
 - Platform Showcase
- 3 Electronics and Software
 - Electronic Design
 - Software Architecture
- 4 Experimental Results
 - Room Scan
 - Transformation Corrections
 - Comparison with a Photograph
- 5 Conclusion
- 6 Odpovědi na otázky

Introduction

Commercially available 3D LiDAR solutions:



Unitree 4D L1

Price: 250 €

Solution: use a cheaper 2D LiDAR and design a stepper-motor-driven platform for it



STL-19P

Price: 80 €

Introduction

Commercially available 3D LiDAR solutions:



Unitree 4D L1

Price: 250 €

Solution: use a cheaper 2D LiDAR and design a stepper-motor-driven platform for it



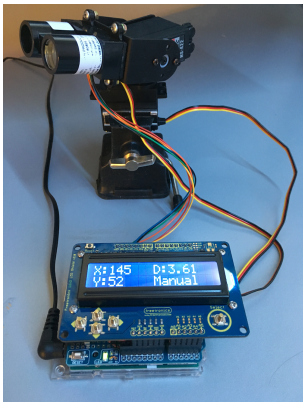
STL-19P

Price: 80 €

Table of Contents

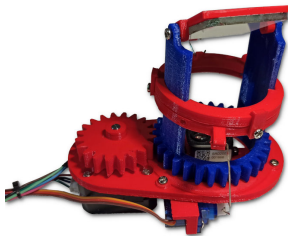
- 1 Introduction
- 2 Platform Design
 - Existing Amateur Solutions for 3D Scanning
 - Point Cloud Visualisation at Different Tilt Angles
 - Dependence of Scan Spacing S_{scan} on Tilt Angle θ
 - Effect of Reduction Ratio R on Scan Spacing S_{scan}
 - Platform Showcase
- 3 Electronics and Software
 - Electronic Design
 - Software Architecture
- 4 Experimental Results
 - Room Scan
 - Transformation Corrections
 - Comparison with a Photograph
- 5 Conclusion
- 6 Odpovědi na otázky

Existing Amateur Solutions for 3D Scanning



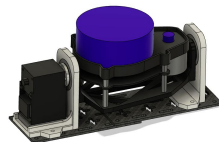
Single-Point LiDAR + Platform

- + Low price
- **Slow scanning**



Single-Point LiDAR + Moving Mirror

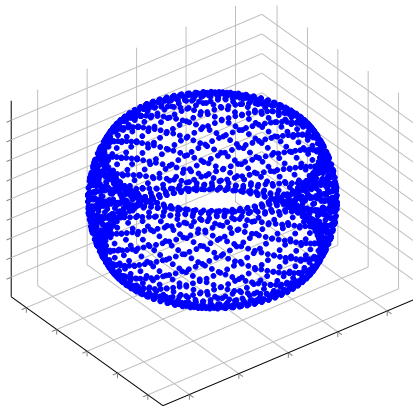
- + Low price
- + LiDAR is stationary (easier to correct the distance measurement)
- High quality mirror is needed
- **Vibrations of the structure**



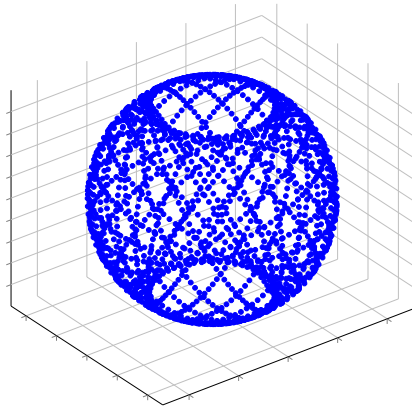
2D LiDAR + Tilting Platform

- + High speed
- + High resolution
- **Uneven point density**

Point Cloud Visualisation at Different Tilt Angles



(a) $\theta = 30^\circ$

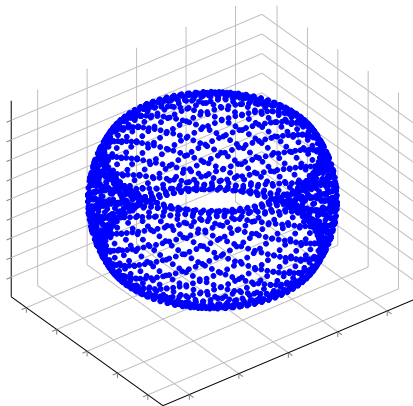


(b) $\theta = 60^\circ$

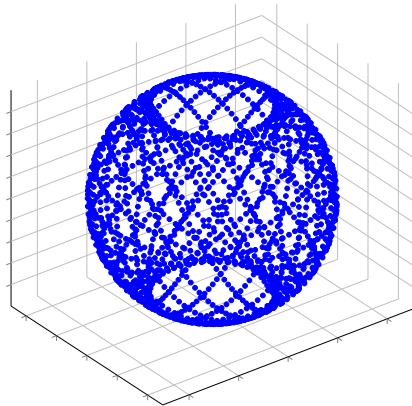
Larger tilt angle θ – more coverage

Problem: Increasing spacing between consecutive LiDAR sweeps as θ increases

Point Cloud Visualisation at Different Tilt Angles



(a) $\theta = 30^\circ$

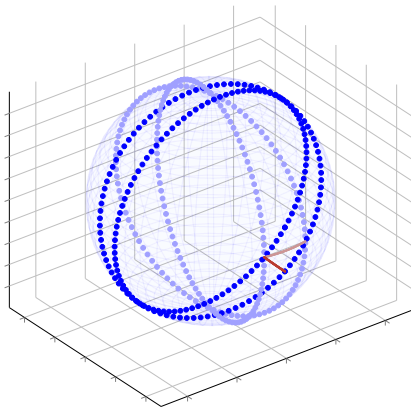


(b) $\theta = 60^\circ$

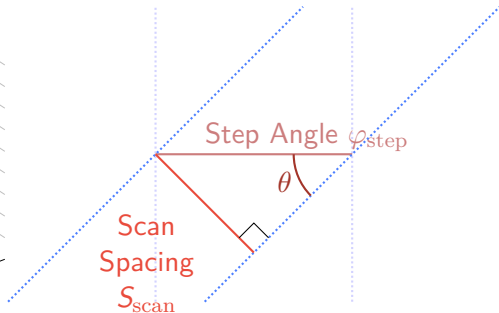
Larger tilt angle θ – more coverage

Problem: Increasing spacing between consecutive LiDAR sweeps as θ increases

Dependence of Scan Spacing S_{scan} on Tilt Angle θ



(a) 3D visualisation



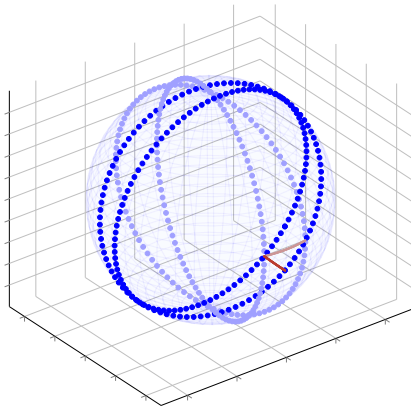
(b) Simplified model showing the dependence

Optimal tilt angle θ : 23.4°

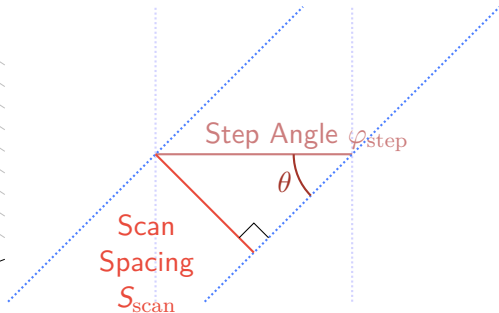
Problem: Limited scene coverage

Solution: Design a gearbox to reduce the effective step angle

Dependence of Scan Spacing S_{scan} on Tilt Angle θ



(a) 3D visualisation



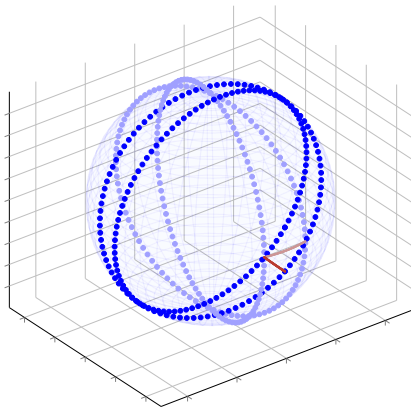
(b) Simplified model showing the dependence

Optimal tilt angle θ : 23.4°

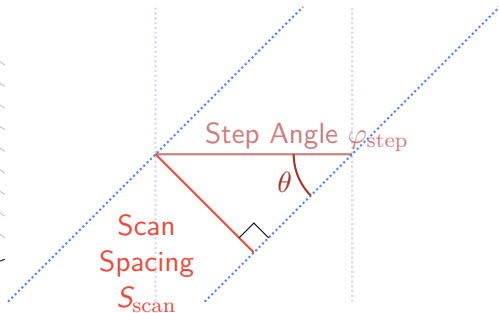
Problem: Limited scene coverage

Solution: Design a gearbox to reduce the effective step angle

Dependence of Scan Spacing S_{scan} on Tilt Angle θ



(a) 3D visualisation



(b) Simplified model showing the dependence

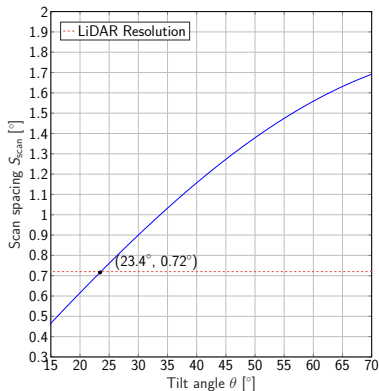
Optimal tilt angle θ : 23.4°

Problem: Limited scene coverage

Solution: Design a gearbox to reduce the effective step angle

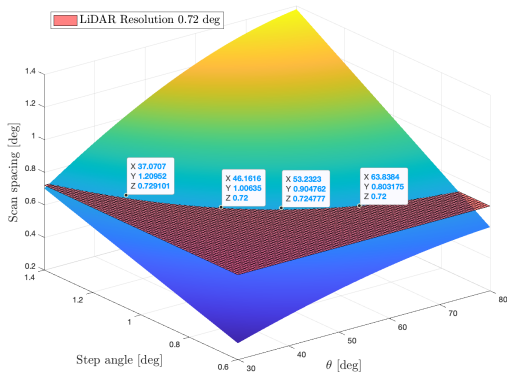
Effect of Reduction Ratio R on Scan Spacing S_{scan}

$$S_{\text{scan}} = \varphi_{\text{step}} \cdot \sin(\theta)$$



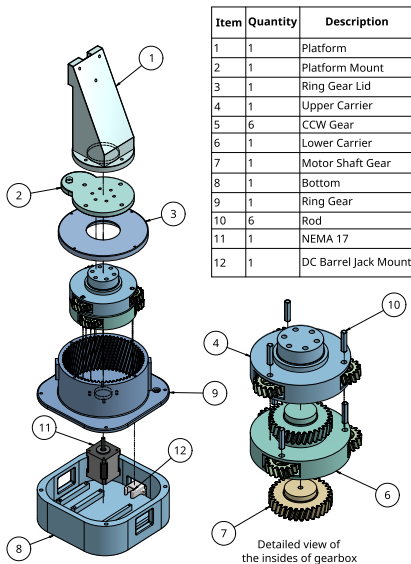
Without a gearbox

$$\varphi_{\text{eff}} = \frac{\varphi_{\text{step}}}{R}, S_{\text{scan}} = \varphi_{\text{eff}} \cdot \sin(\theta)$$



With a gearbox

Platform Showcase



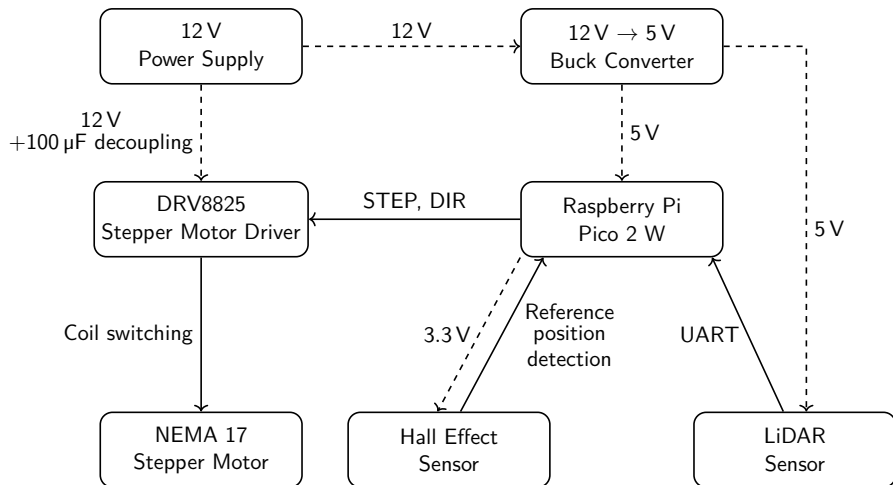
Rigid structure

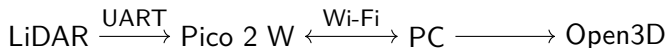
High reduction ratio $R = 18$
($\varphi_{\text{eff}} = 0.1^\circ$) $\rightarrow$ denser point cloud & ability to choose different tilt angles while maintaining the uniform point cloud density

Total price of the platform (with LiDAR and electronics): 164 €

Table of Contents

- 1 Introduction
- 2 Platform Design
 - Existing Amateur Solutions for 3D Scanning
 - Point Cloud Visualisation at Different Tilt Angles
 - Dependence of Scan Spacing S_{scan} on Tilt Angle θ
 - Effect of Reduction Ratio R on Scan Spacing S_{scan}
 - Platform Showcase
- 3 Electronics and Software
 - Electronic Design
 - Software Architecture
- 4 Experimental Results
 - Room Scan
 - Transformation Corrections
 - Comparison with a Photograph
- 5 Conclusion
- 6 Odpovědi na otázky





Pico 2 W

- Homing and motor control
- UART acquisition
- Wi-Fi transmission

PC Receiver Thread

- Packet reception
- 3D reconstruction

PC Main Thread

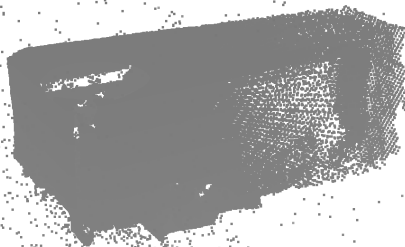
- Open3D visualisation
- Point cloud export

Table of Contents

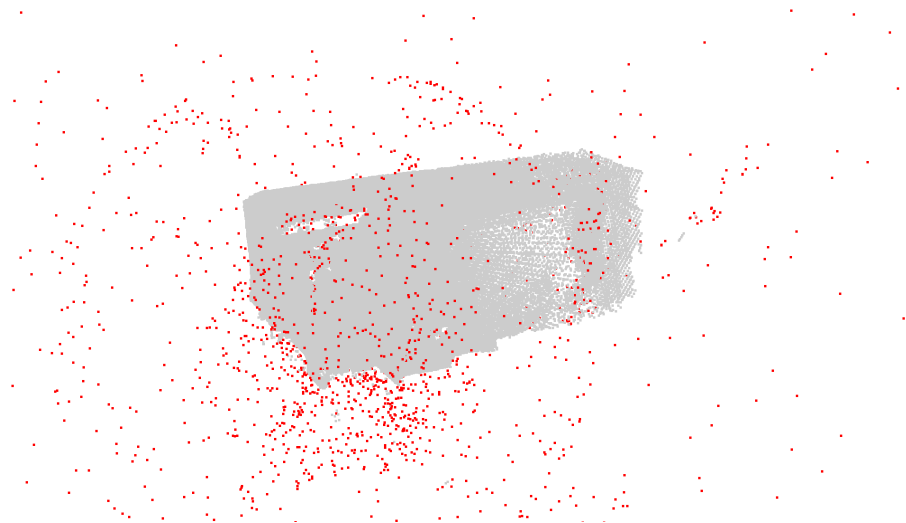
- 1 Introduction
- 2 Platform Design
 - Existing Amateur Solutions for 3D Scanning
 - Point Cloud Visualisation at Different Tilt Angles
 - Dependence of Scan Spacing S_{scan} on Tilt Angle θ
 - Effect of Reduction Ratio R on Scan Spacing S_{scan}
 - Platform Showcase
- 3 Electronics and Software
 - Electronic Design
 - Software Architecture
- 4 Experimental Results
 - Room Scan
 - Transformation Corrections
 - Comparison with a Photograph
- 5 Conclusion
- 6 Odpovědi na otázky

Room Scan

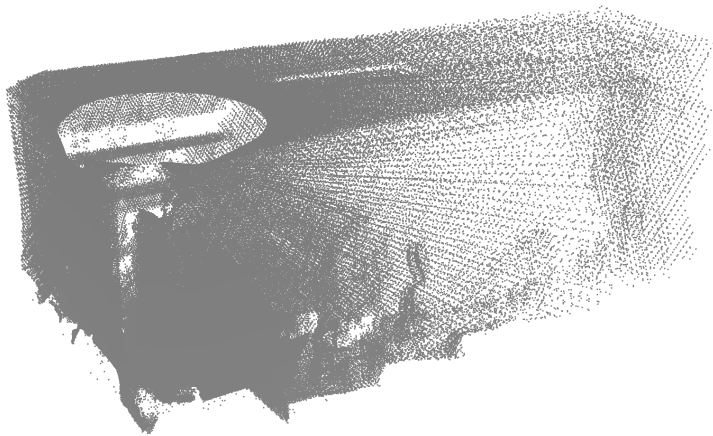
Tested with $\theta = 64^\circ$ and 0.8° step angle between LiDAR sweeps.



Raw point cloud

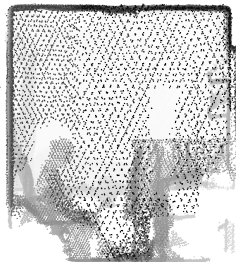
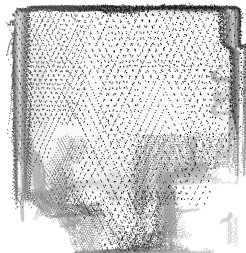


Detected outliers



Filtered point cloud

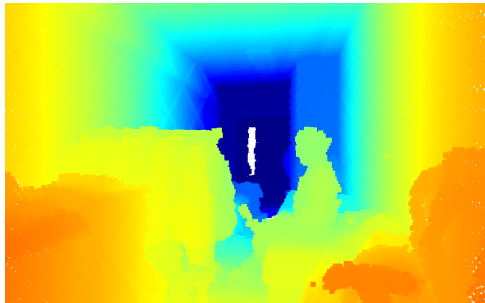
Transformation Corrections



Comparison with a Photograph



Photo of a room



Resulting point cloud

Table of Contents

- 1 Introduction
- 2 Platform Design
 - Existing Amateur Solutions for 3D Scanning
 - Point Cloud Visualisation at Different Tilt Angles
 - Dependence of Scan Spacing S_{scan} on Tilt Angle θ
 - Effect of Reduction Ratio R on Scan Spacing S_{scan}
 - Platform Showcase
- 3 Electronics and Software
 - Electronic Design
 - Software Architecture
- 4 Experimental Results
 - Room Scan
 - Transformation Corrections
 - Comparison with a Photograph
- 5 Conclusion
- 6 Odpovědi na otázky

A platform enabling 3D reconstruction with 2D LiDAR was built. It features a gearbox with a high reduction ratio, allowing for dense point clouds even at large tilt angles.

Code for data acquisition, processing and 3D reconstruction visualisation was written.

The prototype was tested by scanning a room, transformation corrections were applied.

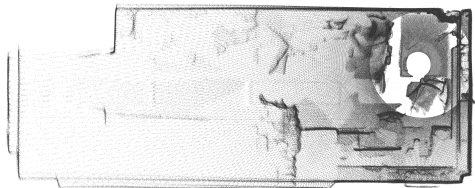
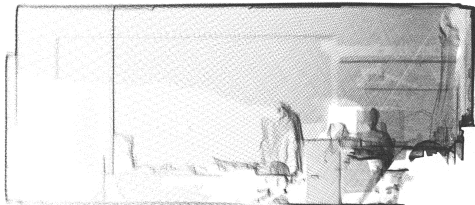
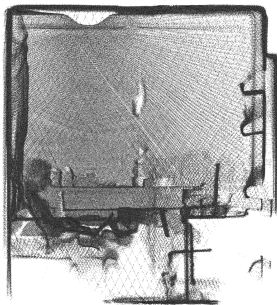
Thank you for your attention!

Table of Contents

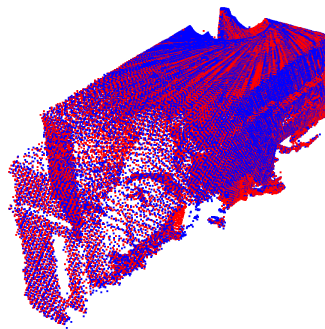
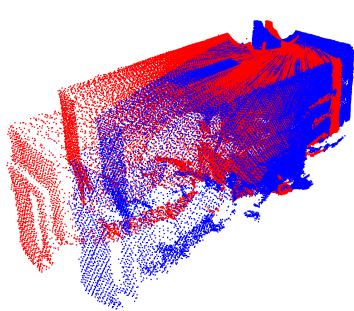
- 1 Introduction
- 2 Platform Design
 - Existing Amateur Solutions for 3D Scanning
 - Point Cloud Visualisation at Different Tilt Angles
 - Dependence of Scan Spacing S_{scan} on Tilt Angle θ
 - Effect of Reduction Ratio R on Scan Spacing S_{scan}
 - Platform Showcase
- 3 Electronics and Software
 - Electronic Design
 - Software Architecture
- 4 Experimental Results
 - Room Scan
 - Transformation Corrections
 - Comparison with a Photograph
- 5 Conclusion
- 6 Odpovědi na otázky

Q1: Jaká je opakovatelnost měření, zejména s ohledem na zarovnání s využitím Hallovy sondy při inicializaci měření?

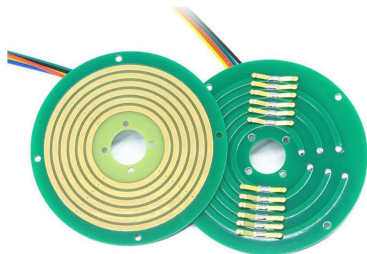
A1.1: Měření bylo opakováno 10krát, point cloudy byly překryty.



A1.2: Open3D nabízí spoustu funkcí pro zarovnávání a porovnávání point cloudů. **Point-to-Point Iterative Closest Point (ICP)** algoritmus byl použit pro demonstrační účely.



Q2: Jaké mechanické úpravy by byly potřeba pro umožnění nepřetržité rotace senzoru? Jakým způsobem by bylo možné kontinuálně scanovat bez nutnosti inicializace orientace před každým spuštěním?



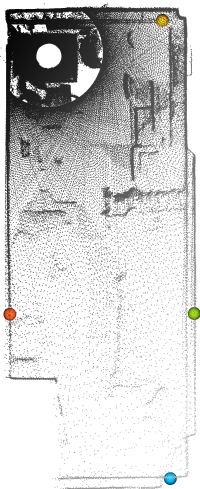
A2: V LiDARech se používají tzv. slip rings pro umožnění přenosu dat a napájení při otáčení o 360° . Neintegrovaná řešení jsou avšak poměrně drahá.

Cena: >100 \$. [Odkaz na Amazon](#)

Q3: Jaká je časová náročnost provedení celého 3D scanu?

A3: $450 \text{ kroků} \cdot 0.105 \text{ s/krok} + 10 \text{ s(homing)} \approx 60 \text{ s}$. Reálná doba je však o cca 10 s delší kvůli přenosu dat přes Wi-Fi. V ojedinělých případech může dojít ke ztrátě spojení nebo komunikačním prodlevám, což může dále prodloužit dobu skenování.

Q4: Odpovídají rozměry nascanovaného pointcloudu reálným rozměrům místnosti? Jaké je relativní zastoupení outlierů v získaném pointcloudu?



Rozměry pokoje: $5.87 \pm 0.05\text{m} \times 2.35 \pm 0.05\text{m}$
(vzato z wiki [Silicon Hill](#))

P0, P1, P2, P3

```
|P1 - P0|: 5.9352 <- length
```

```
|P2 - P1|: 3.0441
```

```
|P3 - P2|: 2.3780 <- width
```

Rozměry nascanovaného pointcloudu jsou v rámci chyby měření

```
Unfiltered pcd: 494400 points
```

```
Outliers: 3737 points
```

```
Filtered pcd: 490663 points
```

```
Removed points percentage: 0.76%
```